

Lab 4: Predicting yearly spending from browsing

70-467 - Machine Learning for Business Analytics

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Background. In class, ridge and lasso were fit on the Credit data, where there are only a handful of predictors. This lab uses a different set of files. An advertising platform has yearly online spending and web browsing for 10,000 anonymous households.

Spending is one row per household: an id and total online spending for the year, in dollars. The median is \$510, the mean is \$1,946, and the largest value is \$401,338. The smallest spend is \$1. No household spent zero.

Browsing is a second file, in long form: 2,271,684 rows. Each row is a household, a site code, and a visit count. Site codes run from 1 to 1,000. The names are a third file, one domain per line, in that same order. These are ordinary websites people browse, not pages on one store. Total page views per household run from 25 to 48,786. The median household has 4,088 page views and a positive count on 219 of the 1,000 sites.

The codebook on the next page describes those three files. Age, income, and purchases are not in them. The 1,000 site columns used in the model are not in them either. You build those columns.

The problem. We want a model that predicts a household's yearly online spending from how its page views are split across these 1,000 sites. There are many sites, so there are many possible models. If we put every site into one least-squares fit, the fit on these 10,000 households can look better as we add sites. That can be overfitting. It does not mean the model will predict well for a new household.

This is the same question as in class: how can we build a good predictive model for spending when there are 1,000 sites, and how can we avoid overfitting?

What we will do in the lab. We will use ridge and the lasso from class to work that question.

First we will join the three files and build one row per household, with a column for each site. The column will be that site's share of the household's page views, not the raw visit count. A count of 10 is a large share for a household with 25 page views and a very small share for a household with about 48,000. After this step, every household's shares sum to 100.

The response is $\log(\text{spend})$. In dollars, the median is \$510 and the largest value is \$401,338, so a few households would dominate a fit on the dollar scale. Every household spent at least \$1.

Read the codebook before class. Bring a laptop.

Codebook. Three files. 10,000 households. The share columns used in the model are not in these files.

browser-totalspend.csv. 10,000 rows, 2 columns. One row is one household.

Variable	Type	Unit	Description
id	integer	none	Household id. The same id is used in the browsing file.
spend	numeric	dollars	Total online spending for the year. The outcome.

browser-domains.csv. 2,271,684 rows, 3 columns. One row is one household and one site.

Variable	Type	Unit	Description
id	integer	none	Household id. Matches id in the spending file.
site	integer	code	Site code, from 1 to 1,000. Matches the line number in the names file.
visits	integer	count	Page views of that site by that household.

browser-sites.txt. 1,000 lines. No header. Line 1 is site code 1.

Variable	Type	Unit	Description
line	text	domain	One website name. The line number is the site code.